

ADITYA ENGINEERING COLLEGE & TECHNOLOGY (A)

Approved by AICTE, Permanently Affiliated to JNTUK & Accredited by NAAC with 'A+' Grade

Recognized by UGC under the sections 2(f) and 12(B) of the UGC act 1956

Aditya Nagar, ADB Road - Surampalem – 533437, E.G.Dist., A.P.,

Department Of CSE - Data Science



CERTIFICATE

This is to certify that the Internship report entitled “**FULL STACK DEVELOPMENT WITH REACT NATIVE**” is being submitted By **CHITTUDI UMA MAHESWARA RAO** Roll no: **22P31A4405**. In partial fulfillment of the requirements for award of the B. Tech degree in CSE – Data Science for the academic year 2025- 2026.

Internship Coordinator

Mrs. Talluri Sushma ,M.Tech,(Ph.D.)
Assistant Professor
CSE-Data Science

Head of the Department

Dr.B.ManikyalaRao,M.Tech.,Ph.D
Associate Professor
CSE-Data Science

ACKNOWLEDGEMENT

I offer my sincere thanks to precious and dynamic Principal, **Dr. A. Ramesh**, M.Tech.,Ph.D, **Aditya College of Engineering & Technology (A)** for his co-operation.

My sincere thanks to **Dr.B.ManikyalaRao**,M.Tech.,Ph.D, Head of the department of CSE – Data Science for his valuable support.

I express my sincere gratitude to my internship guide, **Mrs. Talluri Sushma** ,M.Tech,(Ph.D.) **Assistant Professor, Department of CSE-Data Science**, for her valuable guidance and encouragement which has been helpful in successful completion of this internship.

With immense pleasure I would like to express my deep sense and heart full thanks to the management of **Aditya College of Engineering & Technology (A)**.

CHAPTER 4: WEEKLY REPORTS

4.1 Activity Log for the First Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-06-02	Orientation for the practicum; introduction to Full Stack Development and the MERN stack.	Comprehension of practicum objectives and the functional roles of MongoDB, Express, React, and Node.js was established.	
2025-06-03	Configuration of the development environment, including Node.js, npm, VS Code, and Git.	All core development tools were proficiently installed and configured.	
2025-06-04	Introduction to Node.js; construction of a basic web server utilizing the http module.	The fundamental principles of Node.js and its event-driven architecture were assimilated.	
2025-06-05	Introduction to the Express.js framework, encompassing server setup and basic routing.	The ability to instantiate a simple Express.js server and handle GET requests was acquired.	
2025-06-06	In-depth examination of Express middleware and the request-response lifecycle.	An understanding of how to employ middleware for logging, parsing, and Cross-Origin Resource Sharing was developed.	
2025-06-07	Review of Week 1 activities; establishment of a project repository on GitHub.	Learning was consolidated, and proficiency with fundamental Git commands was achieved.	

4.2 Activity Log for the Second Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-06-09	Introduction to NoSQL databases; configuration of a MongoDB Atlas cluster.	An understanding of the advantages of NoSQL was gained, and a cloud database was successfully provisioned.	
2025-06-10	Introduction to Mongoose; connecting to MongoDB and defining data schemas.	The ability to connect a Node.js application to MongoDB and define data structures via schemas was acquired.	
2025-06-11	Implementation of Mongoose Models; execution of CRUD operations (Create & Read).	API endpoints were authored to facilitate the creation and retrieval of data from the database.	
2025-06-12	Implementation of CRUD operations (Update & Delete).	The backend API was completed, endowing it with full data manipulation capabilities.	
2025-06-13	Structuring Express routes for modularity using express.Router().	The technique of organizing API endpoints into discrete files for enhanced maintainability was learned.	
2025-06-14	Comprehensive testing of all backend API endpoints utilizing Postman.	Proficiency was gained in the domain of API testing and the debugging of server responses.	

4.3 Activity Log for the Third Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-06-16	Introduction to React; core concepts including Components, JSX, and Props.	The fundamental architectural constructs of a React application were understood.	
2025-06-17	Initialization of a React project using the create-react-app utility.	A new single-page application was successfully bootstrapped.	
2025-06-18	State management in React utilizing the useState Hook.	The methodology for managing component-level state and precipitating re-renders was learned.	
2025-06-19	Handling of events and forms within the React paradigm.	The ability to construct interactive forms and manage user input was acquired.	
2025-06-20	Conditional rendering and the rendering of data lists.	Proficiency was achieved in the dynamic display of UI elements based on application state.	
2025-06-21	Review of Week 3 activities; construction of a simple single-page user interface.	React concepts were consolidated through the construction of a small, self-contained frontend application.	

4.4 Activity Log for the Fourth Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-06-23	Utilization of the useEffect Hook for handling side effects (e.g., data fetching).	The method for fetching data from an API upon component mounting was learned.	
2025-06-24	Connection of the React frontend to the backend API utilizing the Axios library.	Data was successfully fetched from the Express API and displayed within the user interface.	
2025-06-25	Implementation of client-side routing with the React Router library.	The ability to simulate a multi-page experience within a single-page application was acquired.	
2025-06-26	Global state management utilizing the Context API and preventing common vulnerabilities	An understanding was gained of how to share state, such as user authentication status, across the application.	
2025-06-27	Full MERN stack integration: Connection of registration and login forms to the API.	A complete authentication workflow was constructed, spanning from the frontend to the backend.	
2025-06-28	Styling of the React application with CSS and a component library (e.g., Material-UI).	The techniques for creating a visually appealing and professional user interface were learned.	

4.5 Activity Log for the Fifth Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-06-30	Introduction to Mobile Development and the React Native framework.	An understanding of the benefits of cross-platform development with React Native was gained.	
2025-07-01	Configuration of the React Native environment utilizing the Expo CLI.	Expo was successfully installed, and a basic application was executed on a device or emulator.	
2025-07-02	Study of Core Components: <View>, <Text>, <Image>, <TextInput>.	Knowledge of the fundamental building blocks for constructing a mobile user interface was acquired.	
2025-07-03	Styling in React Native with the StyleSheet API.	The ability to apply styles to components through the use of JavaScript objects was developed.	
2025-07-04	Layout implementation with Flexbox in React Native.	Mastery of Flexbox was achieved for the purpose of creating responsive layouts adaptable to various screen sizes.	
2025-07-05	Review of Week 5 activities; construction of a simple static mobile screen.	Core React Native concepts were consolidated through the construction of a single-screen user interface.	

4.6 Activity Log for the Sixth Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-07-07	Handling touch events with the <TouchableOpacity> and <Button> components.	The methods for endowing a mobile application with interactivity were learned.	
2025-07-08	Construction of data lists with the <FlatList> and <ScrollView> components.	An understanding was gained of how to efficiently render long, scrollable lists of data.	
2025-07-09	Introduction to navigation utilizing the React Navigation library.	The basic principles of navigating between screens within a mobile application were learned.	
2025-07-10	Implementation of a Stack Navigator for managing screen transitions.	The ability to create a fundamental navigation flow, wherein screens are pushed onto and popped from a stack, was acquired.	
2025-07-11	Construction and validation of forms using the <TextInput> component.	The techniques for creating forms to capture user input in a mobile context were learned.	
2025-07-12	Connection of the React Native application to the MERN backend API.	Data was successfully fetched from the existing API and rendered within the mobile application.	

4.7 Activity Log for the Seventh Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-07-14	Implementation of a Tab Navigator for primary application sections.	A different style of navigation, common in mobile applications, was learned and implemented.	
2025-07-15	Passing data between screens through the use of route parameters.	An understanding of how to transmit information to a new screen during a navigational transition was acquired.	
2025-07-16	Introduction to the Expo SDK for accessing native device APIs.	The utility of the Expo framework for accessing device features without requiring ejection was learned.	
2025-07-17	Utilization of the device Camera via the expo-camera module.	A feature was implemented to capture a photograph using the device's integrated camera.	
2025-07-18	Retrieval of device location coordinates via the expo-location module.	A feature was implemented to fetch and display the user's current geographical coordinates.	
2025-07-19	Secure local data storage on the device using the expo-secure-store module.	The methodology for persisting sensitive data, such as authentication tokens, was learned.	

4.8 Activity Log for the Eighth Week

Day & Date	Brief Description of the Daily Activity	Learning Outcome	Signature
2025-07-21	Finalization of the mobile application's UI/UX; ensuring design consistency.	The application's aesthetic and user experience were enhanced.	
2025-07-22	End-to-end testing of the full stack application, including web and mobile clients.	Comprehensive testing was performed to identify and remediate software defects.	
2025-07-23	Preparation for deployment: Creation of a production build for the React web application.	The process of bundling and optimizing a web application for deployment was learned.	
2025-07-24	Building a standalone mobile application binary using the eas build command.	An understanding of the process for creating a distributable binary file for app stores was gained.	
2025-07-25	Preparation of the final project documentation.	Detailed documentation was authored, explaining the project's architecture and features.	
2025-07-26	Preparing the final project documentation and presentation	A presentation was created to showcase the project and summarize the learning outcomes of the practicum.	